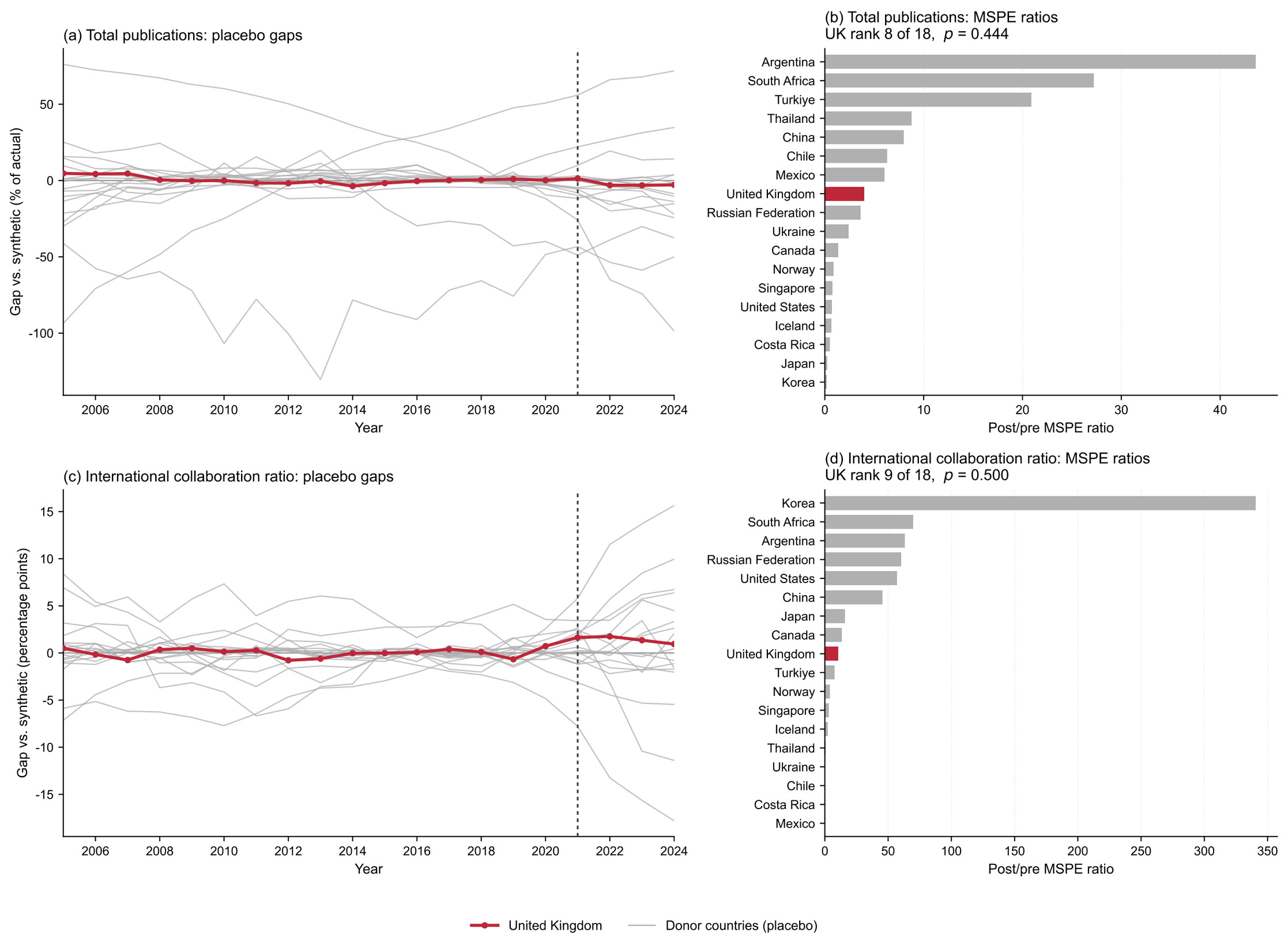
(a) Total publications: placebo gaps
Gap vs. synthetic (% of actual)
50
0
-50
-100
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
Year
(b) Total publications: MSPE ratios
UK rank 8 of 18, p = 0.444
Argentina
South Africa
Turkiye
Thailand
China
Chile
Mexico
United Kingdom
Russian Federation
Ukraine
Canada
Norway
Singapore
United States
Iceland
Costa Rica
Japan
Korea
0
10
20
30
40
Post/pre MSPE ratio
(c) International collaboration ratio: placebo gaps
Gap vs. synthetic (percentage points)
15
10
5
0
-5
-10
-15
2006
2008
2010
2012
2014
2016
2018
2020
2022
2024
Year
(d) International collaboration ratio: MSPE ratios
UK rank 9 of 18, p = 0.500
Korea
South Africa
Argentina
Russian Federation
United States
China
Japan
Canada
United Kingdom
Turkiye
Norway
Singapore
Iceland
Thailand
Ukraine
Chile
Costa Rica
Mexico
0
50
100
150
200
250
300
350
Post/pre MSPE ratio
United Kingdom
Donor countries (placebo)